

Hypothesis Testing Procedure

- **Critical Region C :**
reject H_0 (accept H_1): if $(X_1, \dots, X_n) \in C$
accept H_0 (reject H_1): if $(X_1, \dots, X_n) \in C^c$.
- A **Type I** error occurs if H_0 is rejected when it is true while a **Type II** error occurs if H_0 is accepted when H_1 is true.
- The goal is to select a critical region from all possible critical regions which minimizes the probabilities of these errors. In general, this is impossible. For example, let $C = \emptyset$, type I error is 0 but the type II error is 1.
- Often we consider type I error to be the worse of the two errors. We select the critical region which bounds the type I error and minimizes the type II error.

- We say a critical region C is of **size** α (**significance level**) if

$$\alpha = \max_{\theta \in \omega_0} \mathbb{P}_{\theta}[(X_1, \dots, X_n) \in C].$$

- Over all critical regions of size α , we consider critical regions which have lower probabilities of Type II error.
- For $\theta \in \omega_1$, we want to maximize

$$1 - \mathbb{P}_{\theta}[\text{Type II error}] = \mathbb{P}_{\theta}[(X_1, \dots, X_n) \in C].$$

The probability on the right hand side is called the **Power** of the test at θ .
The **power function** is defined as

$$\begin{aligned}\gamma_C(\theta) &= \mathbb{P}_{\theta}[(X_1, \dots, X_n) \in C] \\ &= \mathbb{P}(\text{accept } H_1 | H_1 \text{ is true})\end{aligned}$$

Testing for a Binomial Proportion of Success

- Let X be a Bernoulli random variable with probability of success p .
Suppose we want to test

$$H_0 : p = p_0 \quad \text{vs.} \quad H_1 : p < p_0.$$

- Let $X_1, \dots, X_n$ be a random sample from the distribution of X and let $S = \sum_{i=1}^n X_i$. An intuitive decision rule (critical region) is
"Reject H_0 in favor of H_1 if $S \leq k$ ",
where k is such that $\alpha = \mathbb{P}_{H_0}[S \leq k]$. Since S is Binomial, we may find k which solve this equation. For example, $n = 20$, $p_0 = 0.7$ and $\alpha = 0.15$, then, $S \sim \text{Bin}(20, 0.7)$ and $k \doteq 11$.

- The power function is

$$\gamma(p) = \mathbb{P}_p[S \leq k], \quad p < p_0.$$

Note that the function is decreasing. The power is higher to detect the alternative $p = 0.2$ than $p = 0.6$.

- **Simple Hypothesis:** completely specifies the underlying distribution, e.g.,
 $H_0 : p = p_0$.
Composite hypotheses: compose of many simple hypothesis, e.g.,
 $H_1 : p < p_0$.

- The test in the last example is based on the exact distribution of its test statistics, i.e., the binomial distribution. Often we cannot get the distribution of test statistics in closed form. Use **central limit theorem**.
- Let X be a random variable with mean μ and variance σ^2 . We want to test

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0,$$

where μ_0 is specified.

- For example: μ_0 is the mean level on a standardized test of students who have been taught by a standard method of teaching.
- Let $X_1, \dots, X_n$ be a random sample from the distribution of X .

- Because $\bar{X} \xrightarrow{P} \mu$, an intuitive decision rule is given by

Reject H_0 in favor of H_1 if $\bar{X}$ is much larger than μ_0 .

- Using the central limit theorem,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \xrightarrow{D} Z.$$

Using this, we obtain a test with an approximate size α , if

$$\text{Reject } H_0 \text{ in favor of } H_1 \text{ if } \frac{\bar{X} - \mu}{S/\sqrt{n}} \geq z_\alpha.$$

- The test is intuitive. To reject H_0 , $\bar{X}$ must exceed μ_0 by at least $z_\alpha S/\sqrt{n}$.
- The power function is approximated by

$$\begin{aligned}\gamma(\mu) &= \mathbb{P}_\mu(\bar{X} \geq \mu_0 + z_\alpha \sigma/\sqrt{n}) \\ &\doteq 1 - \Phi\left(z_\alpha + \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma}\right) \\ &= \Phi\left(-z_\alpha - \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma}\right),\end{aligned}$$

which is an increasing function of μ .

- Let $X \sim N(\mu, \sigma^2)$. Consider

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0.$$

- Under H_0 , the test statistics $T = (\bar{X} - \mu)/(S/\sqrt{n})$ has a t -distribution with $n - 1$ degrees of freedom.
- The decision rule is

$$\text{Reject } H_0 \text{ in favor of } H_1 \text{ if } T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \geq t_{\alpha, n-1}.$$